Chenjie Li

Chicago, IL | (224) 602-1109 | cli206@uic.edu | LinkedIn | chenjie-li-cs.github.io

Research Scientist, Machine Learning (PhD) — ML Data, Weak Supervision & LLM-Assisted Labeling

SUMMARY

Computer Science Ph.D. candidate at the University of Illinois Chicago (expected December 2026; advisor: Prof. Boris Glavic) researching data-centric machine learning, weak supervision, and explainable and causal analytics. Lead author of work published at KDD 2025 and SIGMOD 2021 on refining labeling functions with limited labeled data and explaining query results. Experience developing LLM-assisted labeling pipelines with GPT-4 and Snorkel, and production AI / data pipelines using Python, SQL, PyTorch, and AWS.

Research interests: Weak supervision and labeling functions; LLM-assisted data labeling and data-centric ML; explainable analytics; causal inference for relational data; semantic type detection for structured data.

EDUCATION

University of Illinois Chicago — Expected Dec. 2026
Ph.D. in Computer Science — Advisor: Prof. Boris Glavic

Illinois Institute of Technology — 2018–2024
M.S. in Data Science; Ph.D. studies (transferred)

Guizhou University — 2013–2017
B.S. in Electrical Engineering

SELECTED PUBLICATIONS

C. Li, A. Gilad, B. Glavic, Z. Miao, S. Roy. Refining Labeling Functions with Limited Labeled Data. **KDD 2025**.

C. Li, D. Zhang, J. Wang. LLM-Assisted Labeling Function Generation for Semantic Type Detection. **VLDB DATAI Workshop 2024**.

C. Li, J. Lee, Z. Miao, B. Glavic, S. Roy. CaJaDE: Explaining Query Results by Augmenting Provenance with Context. **PVLDB Demo**.

C. Li, Z. Miao, Q. Zeng, B. Glavic, S. Roy. Putting Things into Context: Rich Explanations for Query Answers Using Join Graphs. **SIGMOD 2021**.

RESEARCH EXPERIENCE

University of Illinois Chicago — Research Assistant — Aug. 2024 – Present
Advisor: Prof. Boris Glavic | Chicago, IL

RuleCleaner — KDD 2025, lead author

- Led technical development of a framework for repairing noisy weak-supervision labeling functions with limited labeled data; designed optimization-based (MILP) and heuristic algorithms to refine labeling rules while maintaining interpretable outputs.
- Improved global labeling accuracy on **8 of 9 datasets** using **at most 150 labeled examples**; evaluated on **11 real datasets up to 200K rows** (Amazon, 200,000 rows). On ChemProt, **RuleCleaner + Snorkel reached 37.9% vs GPT-4o at 27.4%**. Ran **56x to 1,312x faster than Llama-3-8B-Instruct** as a labeler (about 2-4 orders of magnitude faster).

RelCausal — in progress

- Developing a relational causal explanation system for investigating aggregate SQL query results using causal DAGs, treatment-effect estimation, and scalable search.

Illinois Institute of Technology — Research Assistant Jan. 2020 – May 2024
Chicago, IL

- Developed **CaJaDE**, an explainable data analytics system that investigates unexpected SQL query results using contextual information from relational databases; implemented query explanation algorithms and an interactive prototype.
- Evaluated via case studies on **2 real-world datasets** — NBA (10 seasons, 2009-10 to 2018-19, 11 relations, ~170MB) and MIMIC ICU (~120MB, 6 relations) — with **5 queries per dataset, 10 queries total**.
- User study with **20 participants**: **16 of 20** agreed CaJaDE explanations make more sense and preferred CaJaDE over provenance-only explanations; ranking quality (NDCG) reached **0.90 and 0.95** after dropping the most controversial explanation.
- Showed **sublinear scaling** on datasets scaled from ~10% to ~800% (scale factor 8); sampling at 0.1 was **>55% faster** than 0.7 at scale factor 8.
- **~35x faster** than the Explanation Tables (ET) baseline at sample size 512 (**24.51s vs. 855.13s**).

INDUSTRY RESEARCH EXPERIENCE

Megagon Labs — Research Scientist Intern Jan. – Apr. 2024
Mountain View, CA

- Developed an end-to-end LLM-assisted data labeling pipeline using GPT-4 and few-shot prompting to generate labeling functions for semantic type detection.
- Implemented rule-based labeling functions using keywords, regular expressions, and statistical features; integrated Snorkel to aggregate noisy predictions into pseudo-labeled training data.
- Built and evaluated a stacked weak-supervision architecture for large-scale structured data, supporting datasets with up to 570K tables and 255 semantic types.

CCC Intelligent Solutions — Data Science Engineer Intern May – Dec. 2025; Sept. 2026 – Present
Chicago, IL

- Developed an automated video sanitization pipeline for insurance-claim data, integrating Airflow, Kubernetes, AWS S3, and GPU-based ML inference.
- Designed and orchestrated multi-stage workflows covering data ingestion, distributed inference, and automated anonymization of faces and license plates.
- Fine-tuned a YOLOv8 detection model through progressive training on OpenImages and internal claim data, achieving 0.908 mAP50 for face and license-plate detection.
- Developed MySQL-backed Grafana dashboards to monitor pipeline execution, ingestion health, operational metrics, failure patterns, and data-processing outcomes.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C/C++, Bash

AI / ML: LLM prompting, few-shot learning, weak supervision, model evaluation, optimization, causal inference

ML / Data Platforms: PyTorch, scikit-learn, Snorkel, Apache Spark, Airflow, AWS EMR / S3, Databricks

Systems / Tools: Kubernetes, Docker, Linux, Git, MySQL, PostgreSQL, Grafana

RESEARCH FOCUS FOR META

Data-centric ML for large-scale systems: programmatic weak supervision and LLM-generated labeling functions that improve training-data quality with limited labeled data; evaluation of label models and downstream ML performance; and interpretable, causal explanations for complex data and query results.